

IT Infrastructure Plan

Office Space Expansion (2026/2027)
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Approach and Initial Assumptions

Since the floor plan didn't include information about the existing network, I started from a few realistic assumptions that I would confirm at the very start of the work:

- The facility already has basic network infrastructure (an internet connection, at least one switch, likely a local server or NAS) in the existing office area; these 5 new offices connect to that infrastructure, rather than building a network from scratch.
- All offices (existing and new) are on the same floor / in the same building, so no fiber connection between buildings is needed; only local cabling is required.
- I read the number of workstations directly from the floor plan: 4 desks with a computer (rooms K1-K3) and 2 meeting rooms (K4 and K5) using a laptop setup; the equipment plan is sized accordingly, not based on an assumed headcount.

Note: the diagram and attached layout are entirely based on the actual floor plan I received, with the exact number of rooms, doors, and desks. It is not a generic template. All devices are precisely centered on the desks and kept clear of entry doors for safety and aesthetics.

1. Equipment Layout

The layout was built directly on top of the provided floor plan, to show exactly where each device goes and how cables run from each outlet to the central communications cabinet (RACK).

Brief description by room:

- **K1 - Single office:** 1 workstation (PC 1) + IP phone positioned exactly at the center of the yellow desk. The existing meeting table remains without fixed IT equipment.
- **K2 - Single office:** 1 workstation (PC 2) + IP phone centered on the blue desk. The central RACK cabinet is mounted on the side wall, well clear of the entry door.
- **K3 - Shared office with two blue desks:** 2 workstations (PC 3, PC 4) + 2 IP phones positioned exactly at the center of each desk. The shared network printer sits in an optimal spot next to desk 4.
- **K4 and K5 - Meeting rooms:** instead of classic fixed computers, dedicated USB-C multi-port HUB units are mounted at the center of the large blue tables for quickly connecting a laptop to all peripherals. Large presentation TV screens are wall-mounted, well clear of entry doors for better visibility and to avoid damage when doors open.
- **RACK (Communications cabinet):** placed in K2, at a safe distance from the door, giving a shorter network uplink and easier ongoing maintenance.
- **AP1 and AP2 (WiFi Access Points):** strategically positioned on the wall or ceiling in the lower zone of the rooms, well clear of entry doors, achieving ideal coverage and eliminating dead zones.

2. Equipment List with Specifications

Item	Qty	Approximate specification	Purpose / note
Desktop computer (PC)	4	Intel Core i3/i5 (12th gen or newer), 8-16 GB RAM, 256-512 GB SSD, Windows 11 Pro	Workstations K1-K3, perfectly centered on desks.
Monitor	4	21.5-24", Full HD, HDMI/DP	One per each fixed PC for stable daily use.
IP phone (VoIP)	4	PoE, 2 network ports (pass-through)	Telephony runs over the network cable alongside the computer at the center of the desk.
USB-C Multi-port HUB	2	Built-in desktop hub with HDMI, USB-A, USB-C and RJ45 ports	Meeting rooms K4 and K5 - placed at the center of the blue tables for instant laptop connection to the network and TV.
Meeting room screen (TV)	2	43-50", Smart TV, HDMI + USB-C, sturdy wall mount	Mounted on the walls in K4 and K5, at a safe distance from the door.
Network printer (MFP)	1	Laser, duplex, network (Ethernet + WiFi)	Shared multifunction device for all new offices, shared over the local network.
WiFi Access Point	2	PoE, dual-band (WiFi 6), centrally manageable (cloud/controller)	AP1 and AP2 for stable coverage on the wall/ceiling in the lower section, away from the door.
Access switch (RACK)	1	24-port, Gigabit, PoE+ (min. 8 PoE ports), managed/VLAN	Central switch in the new RACK cabinet, uplinked to the existing core switch.
Small wall rack cabinet	1	9-12U, with built-in ventilation and locking	Secure housing for the switch, patch panel, and UPS unit in K2.
Patch panel + modules	1 set	24-port Cat6 patch panel + RJ45 Cat6 keystone jacks	Clean termination of network lines from all offices in the RACK cabinet.
Cat6 UTP cable	~150 m	Cat6 U/UTP, indoor, 100% copper	Reliable network runs following the lower wall route directly to the RACK.
UPS (uninterruptible power supply)	1	650-1000 VA, line-interactive with stabilization	Protects the switch, router, and network links from power surges and outages.
Cable trunking / trays	as needed	PVC wall and floor cable trunking	Neat, concealed cable routing along walls and floors to the outlets.

3. Proposed Network Topology

I propose a proven star topology that connects to the facility's existing network structure via one main uplink cable. This solution offers simplicity for future expansion and easy troubleshooting of potential faults.

- **Star topology:** each device has its own independent Cat6 UTP cable running to the Access Switch in the RACK; a break in one cable does not affect the rest of the network.
- **Access Switch:** connected via one fast uplink cable to the existing Core Switch, so there is no need to purchase a new router.
- **Segmentation:** traffic is split into VLAN 10 (workstations and phones), VLAN 20 (WiFi employees), and VLAN 30 (WiFi guests).

4. Security Solutions

Network security:

- **VLAN segmentation:** complete separation of business computers from the visitor network (VLAN 30), preventing unauthorized access to local servers.
- **Firewall policies:** blocking unnecessary inbound traffic from the internet and configuring secure VPN tunnels for remote work.
- **Port Security:** preventing unknown devices from joining the internal network simply by plugging into an open wall outlet.

Data and device protection:

- **Active Directory integration:** centralized management of passwords, access rights, and security patches through group policies (GPO).
- **Backup strategy:** automatic daily backup of all business data to a separate location or protected cloud storage, following the 3-2-1 rule.
- **Physical protection:** the RACK cabinet is permanently locked; access to active equipment is granted only to authorized personnel.

5. Prioritization and Implementation Phases

- **Phase 1 - Passive equipment and cabling:** laying cable runs and trunking along the lower wall, and mounting the RACK cabinet in K2 away from the door. Done first, as it makes every following step easier while the rooms are still unfurnished.
- **Phase 2 - Active network equipment:** mounting the switch, configuring VLANs, and mounting access points (AP1 and AP2) on the wall/ceiling away from the entrance. The network is fully tested before equipment is connected.
- **Phase 3 - End devices:** installing 4 desktop computers exactly at the center of the desks, built-in HUBs in K4 and K5, monitors, IP phones, the shared printer, and 2 TVs on the lower walls. Devices are joined to the corporate domain.
- **Phase 4 - Security and documentation:** activating antivirus licenses and protection policies, a final WiFi coverage check, and producing a complete network diagram with IP addresses.

6. Additional Proposal

I propose introducing a simple digital reservation system for the meeting rooms (K4 and K5), either via a shared Outlook/Google calendar or by mounting a small, affordable tablet on the wall next to the entrance of each room.

Given that the facility has multiple departments and two meeting rooms, this solution completely eliminates scheduling conflicts over meeting times. It represents a minimal investment with an immediate positive impact on work organization.